

Deep Learning Specialization: Course Introduction

1. The Impact of Deep Learning

- **Transformation of Industry:** Deep learning has revolutionized traditional internet services (search, advertising) and is currently enabling new sectors:
- **Healthcare:** Automated reading of X-ray images.
- **Education:** Personalized learning experiences.
- **Agriculture:** Precision farming techniques.
- **Transportation:** Development of self-driving cars.
- **"AI is the New Electricity":** A historical analogy suggesting that just as electrification transformed every major industry 100 years ago, AI is now on a path to bring an equally fundamental transformation to modern society.

2. Specialization Curriculum Overview

The specialization consists of five courses designed to move from foundational theory to advanced application.

Course 1: Neural Networks and Deep Learning (4 Weeks)

- **Focus:** Foundations of neural networks.
- **Key Skills:** Building and training a **Deep Neural Network**.
- **Practical Project:** Building a **cat recognizer** (image classification).

Course 2: Improving Deep Neural Networks (3 Weeks)

- **Focus:** Practical aspects of making models perform well.
- **Key Concepts:** * **Hyperparameter tuning**.
- **Regularization**.
- **Diagnosis of Bias and Variance**.
- **Advanced Optimization Algorithms:** Momentum, RMSprop, and Adam.

Course 3: Structuring Machine Learning Projects (2 Weeks)

- **Focus:** Strategy and best practices for building ML systems.
- **Key Concepts:**
- New methods for data splitting: **Train, Development (Dev/Hold-out), and Test sets**.
- Handling mismatched data distributions.

- **End-to-end Deep Learning:** Understanding when and when not to use it.

Course 4: Convolutional Neural Networks (CNNs)

- **Focus:** Computer Vision applications.
- **Application:** Building models specifically designed for processing image data.

Course 5: Sequence Models

- **Focus:** Processing sequence data and **Natural Language Processing (NLP)**.
- **Key Architectures:**
- **Recurrent Neural Networks (RNNs)**.
- **Long Short-Term Memory (LSTM)** models.
- **Applications:** Speech recognition, music generation, and NLP.

3. Summary of Learning Path

The sequence transitions from **supervised learning** foundations to specialized architectures (CNNs for images, RNNs/LSTMs for sequences) while providing industry-standard strategies for project management.